

Soil Nitrogen Pipeline v3.1 — Model Report

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Summary

v3.1 establishes that a five-year record supports description and association, not prediction — and that the study's only significant management effect belongs to the covariate every importance method ranked last.

The pipeline fuses soil inorganic nitrogen (NH_4^+ , NO_3^-) with Google Earth Engine covariates, a per-window water-balance index, and chamber GHG flux. It ranks covariates three independent ways, fits one global Gaussian Process, and validates it against a held-out season.

What the model supports:

- **Grazing depletes soil N only where no input replaces it.** Unfertilized plots lost 11.48 mg kg^{-1} NH_4^+ and 14.99 mg kg^{-1} NO_3^- under grazing by 2023 (Holm-corrected $p = 0.036$ and 0.030 within the 2023 endpoint family). Fertilized and legume plots showed no significant effect.
- **Grazing ranked last or near-last under every importance method** — 11th of 11 under impurity importance, in the negative-importance group under permutation. A variable can carry a real, conditional effect without being a useful global predictor.
- **Year and water balance lead the rankings.** Both methods agree on year and SPEI for NH_4^+ and NO_3^- . Drier sampling windows coincided with higher inorganic N.

What the model cannot support:

- **Forward prediction.** Trained on 2019–2022 and tested on 2023, R^2 was -3.26 (NH_4^+) and -1.83 (NO_3^-). Credible intervals averaged 53 and 88 mg kg^{-1} wide — wider than the observed range.
- **Treatment rankings.** Treatment carried little global importance, and the model has no validated skill to rank systems.

Study design and data

Nine one-acre plots under three randomized treatments, each split into grazed and ungrazed strips, sampled at two depths from 2019 to 2023.

Treatment	Code	Plots
Unfertilized Triticale	1	102, 202, 302
Fertilized Triticale	2	101, 203, 301
Triticale interseeded with Legume	3	103, 201, 303

Sampling ran at three time points: Time 0 (2019 baseline, before grazing), Time 1 (July), and Time 2 (September). Depths were 0–10 and 10–20 cm. The response dataset (NH4_N03_master) holds 1,363 samples with unique IDs encoding year, plot, time point, depth, grazing condition, and replicate.

After aggregation to plot × year × time point × depth × grazing, the modelling table has 282 rows. Rows lacking GHG covariates (2019, before flux sampling began) drop from the fitted models, leaving 264.

Sample counts are unbalanced by grazing: grazed samples outnumber ungrazed roughly 2:1 in 2021–2022 and about 3.5:1 in 2020. Legume strips were not sampled in 2020.

Data audit and coverage triage

Soil inorganic nitrogen is the only variable measured in all five years, across all treatments, at every time point — so it anchors the model by coverage, not by choice.

Variable	2019	2020	2021	2022	2023	Role
$\text{NH}_4^+ / \text{NO}_3^-$	All	All	All	All	All	Response
GHG flux (CO_2 , N_2O , CH_4)	—	All	All	All	All	Covariate
pH / EC / bulk density	All	Plot IDs lost	—	Time 2 only	Time 2 only	Baseline
Carbon fractions (MAOM, FPOM, DOC)	LOI only	—	—	Time 1 only	Time 2 only	Snapshot
FAME biomarkers	All	—	—	Time 2 only	Time 2 only	Snapshot

"Time 2 only" means all three treatments, second sampling only. Those variables support a treatment comparison within 2022–2023 but cannot form a trajectory.

The 2020 pH and EC records are complete measurements whose plot identifiers were lost. They cannot be assigned to treatments and are excluded entirely — the one case in the dataset that supports nothing.

The carbon fraction and FAME treatment comparison is available but out of scope for this report.

Covariates and data fusion

Four covariate streams join the nitrogen data, each keyed to the finest resolution it genuinely supports.

Stream	Variables	Source	Resolution	Merge key
Vegetation	NDVI, EVI	Sentinel-2 SR via Google Earth Engine	10 m, 15 m buffer per plot	plot × year × time point
Soil moisture	SAR VV	Sentinel-1 GRD via Google Earth	10 m	plot × year × time

Stream	Variables	Source	Resolution	Merge key
proxy	backscatter	Engine		point
Water balance	SPEI-90d	GRIDMET/DROUG HT	~4.6 km (site- level)	year × time point
Greenhouse gas flux	CO ₂ , N ₂ O, CH ₄	Chamber measurements, Hutchinson– Mosier	Per plot, per grazing strip	plot × year × grazing

SPEI. Extracted per sampling window rather than as one annual value. This breaks SPEI's collinearity with year: correlation with year fell to $r = -0.24$ (spei90d), and the index now varies between Time 1 and Time 2. The 4.6 km grid covers the whole field in one pixel, so SPEI cannot vary between plots. GEE values agree in direction with the annual values pulled manually from the SPEI Global Drought Monitor: 2019 is the only wet year in both.

GHG. Chamber fluxes (4–15 dates per year) average to a seasonal mean per plot, year, and grazing strip. Grazing is part of the merge key because flux was measured separately on grazed and ungrazed strips. Outlier CH₄ values above ~50 g ha⁻¹ d⁻¹ (5 rows) and two values near -71 were blanked as physiologically implausible for upland soils; large CO₂ values were kept as plausible peak-season respiration.

Structural covariates. Treatment (one-hot in GPR), grazing (binary), and depth (numeric midpoint) enter as model inputs, not as filters.

Model architecture

Three independent importance methods rank the covariates; one global Gaussian Process fits the trajectory; a temporal holdout tests whether any of it generalizes.

flowchart LR

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A[Soil N + covariates<br/>282 rows] --> B[RF impurity<br/>importance]
A --> C[Permutation<br/>importance]
B --> D[Top covariates]
D --> E[Global GPR<br/>ARD kernel]
E --> F[ARD length-scale<br/>relevance]
E --> G[Temporal holdout<br/>train 2019–22, test 2023]
A --> H[Mann–Whitney U<br/>grazed vs ungrazed]

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The Mann–Whitney branch runs on raw measurements, independent of every model.

Random Forest impurity importance (MDI). 500 trees, `max_features='sqrt'`, `min_samples_leaf=3`. Importance = total variance reduction credited to each covariate, normalized to sum to 1. Evaluated with leave-one-year-out cross-validation.

Permutation importance. One year held out via `GroupShuffleSplit` (grouped by year). Each covariate is shuffled 50 times; importance = the resulting increase in held-out MSE. Negative values mean shuffling improved prediction.

Global GPR. One model over all rows — replacing v3.0's 18 per-subset fits, in which treatment, grazing, and depth were constants and could carry no effect. Target is $\log(1 + N)$. Kernel:

$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \cdot \mathrm{Matérn}_{\nu=1.5}(\mathbf{x}, \mathbf{x}'; \ell_1, \dots, \ell_d) + \sigma_n^2 \delta_{\mathbf{x}\mathbf{x}'}$$

The Matérn term carries one length-scale per input dimension (ARD). Relevance is taken as the inverse length-scale, normalized to sum to 1. Inputs: RF-selected continuous covariates, one-hot treatment, grazing, depth. NH_4^+ used 10 dimensions, NO_3^- 9, both on $n = 264$.

Temporal holdout. Fit on 2019–2022 ($n = 192$), predict 2023 ($n = 72$). Report RMSE, MAE, bias, R^2 , 95% credible-interval coverage, and mean interval width.

Projection assumptions (2024–2028). Treatments continue unchanged. SPEI held at its observed mean (about -0.6). GHG held at 2023 means. NDVI, EVI, and SAR held at training means.

Results: covariate rankings

Year and water balance rank highest under both impurity importance and ARD; grazing ranks at or near the bottom under all three methods.

Covariate	NH_4^+ MDI	NH_4^+ permutation ($\Delta\mathrm{MSE}$)	NO_3^- MDI	NO_3^- permutation ($\Delta\mathrm{MSE}$)	NO_3^- ARD
Year	0.239	0.0	0.133	0.0	0.152
SPEI-90d	0.159	0.0	0.174	0.0	0.198
NDVI	0.138	+1.4	0.104	+0.3	0.320
CO_2 flux	0.155	−7.0	0.109	−2.5	0.149
Depth	0.019	+3.3	0.108	+37.6	0.040
Treatment	0.020	+2.9	0.027	≈ 0	0.061
Grazing	0.004	−0.3	0.008	< 0	0.035

MDI and ARD are shares summing to 1. Permutation is the rise in held-out MSE when a covariate is shuffled. ARD treatment is its largest one-hot term. For NH_4^+ , ARD ranks NDVI, SPEI, and year highest.

Where the methods agree. Covariates in the top four under both MDI and ARD: year, SPEI, and NDVI for NH_4^+ ; SPEI, year, and CO_2 for NO_3^- .

Where they disagree, and why.

- **Year and SPEI score zero under permutation.** With one year held out, year is constant in the test set and SPEI takes only two values. Permutation cannot test them — this is a design limit, not evidence of irrelevance.
- **CO_2 flux degrades held-out prediction.** It ranks 3rd under MDI for both analytes, yet shuffling it improves prediction (−7.0 for NH_4^+). The forest learned a CO_2 –nitrogen pattern that does not transfer across years.
- **Depth is the one covariate both methods rank highly for NO_3^-** (MDI 4th, permutation 1st at +37.6). Consistent with nitrate leaching and ammonium binding to exchange sites.

- **NDVI dominates ARD while EVI nearly vanishes.** The two measure nearly the same thing; ARD concentrates weight on one of a correlated pair where RF splits it.

Direction of the SPEI effect. Importance is unsigned. Lining values up against observed N shows the relationship is inverse: 2019 was the only wet window and had low N; the driest window (2022, Time 2, $\text{spei90d} = -1.81$) coincided with rising N. SPEI still correlates with year at $r = -0.24$, so drought and elapsed time are not fully separable.

Results: the grazing conditional effect

By 2023, grazing depleted both nitrogen forms in unfertilized plots and had no significant effect where fertilizer or legume nitrogen was present.

Grazed minus ungrazed, mg kg^{-1} . Mann–Whitney U, two-sided. * $p < 0.05$, ** $p < 0.01$.

Treatment · analyte	2020	2021	2022	2023
Unfertilized · NH_4^+	+1.06	−0.17	−2.63	−11.48**
Unfertilized · NO_3^-	+4.05	+2.94*	−2.86	−14.99**
Fertilized · NH_4^+	−1.85*	−2.88	+1.11	+5.34
Fertilized · NO_3^-	−2.51	+4.73*	−1.48	+7.47
Legume · NH_4^+	—	+0.12	−4.10	+4.71
Legume · NO_3^-	—	+1.79	−9.90	+3.28

The two 2023 unfertilized effects: NH_4^+ $p = 0.0071$, NO_3^- $p = 0.0050$, both on $n = 50$ grazed / 28 ungrazed. Legume strips were not sampled in 2020. Sample counts range from 102–106 / 30 (2020) to 50–70 / 28–38 (2023). 2019 is excluded as the pre-grazing baseline.

Why the importance methods missed it. Global importance asks whether grazing helps explain nitrogen across all plots, years, and treatments at once. An effect confined to one treatment in one year cannot split the whole dataset well — and grazing is binary, which impurity importance additionally penalizes (Strobl et al., 2007). The pairwise test asks a narrower question, and there the effect is clear.

The global GPR agrees. With grazing as a model dimension, its ARD relevance is 0.035 for NO_3^- , and projected grazed and ungrazed trajectories converge in every treatment. v3.0's visibly separated lines came from fitting different subsets, not from a modelled grazing effect.

Multiple comparisons. The table above reports 22 uncorrected tests. This analysis treats 2023 — the endpoint after four grazing seasons — as the primary contrast, and corrects within that family of six tests (3 treatments $\times$ 2 analytes) using the Holm step-down method.

Contrast	Uncorrected p	Holm-corrected p	At $\alpha = 0.05$
NO_3^- , unfertilized, 2023	0.0050	0.030	Significant

Contrast	Uncorrected p	Holm-corrected p	At $\alpha = 0.05$
NH ₄ ⁺ , unfertilized, 2023	0.0071	0.036	Significant

Across all 22 cells the threshold falls to roughly 0.002 under either Bonferroni or Benjamini–Hochberg, and neither result clears it. The choice of family therefore decides the outcome.

The result holds if 2023 — the endpoint after four grazing seasons — is treated as the primary contrast, and does not hold across all 22 cells. That choice was not pre-specified, so it should be stated explicitly rather than implied by an uncorrected $p < 0.01$. The isolated 2020–2021 significant cells run in both directions and survive no correction; treat them as noise. The corrected values are computed as `p_holm_2023` in the dashboard export and confirmed against a notebook run.

Results: temporal holdout validation

Trained on 2019–2022 and tested on 2023, the global GPR predicted worse than the test-year mean for both analytes.

Metric	NH ₄ ⁺	NO ₃ [−]
Train / test rows	192 / 72	192 / 72
RMSE (mg kg ^{−1})	22.28	17.52
R ²	−3.26	−1.83
95% credible-interval coverage	100%	100%
Mean interval width (mg kg ^{−1})	53.1	88.3
RF leave-one-year-out R ² (mean)	−2.53	−1.13

Coverage of 100% is not a success here. The intervals are wider than the full observed range of either analyte, so every observation falls inside them trivially. Near-95% coverage with intervals narrower than the data spread would indicate calibrated uncertainty; this is the opposite — uninformative uncertainty. Coverage has to be read alongside width.

Why predictions flatten. Predicting 2023 places year outside the training range. A Matérn GP extrapolating beyond its data reverts toward the prior mean, so held-out NH₄⁺ predictions collapse to a near-constant value regardless of plot, treatment, or grazing. Roberts et al. (2017) describe exactly this: blocking in time forces extrapolation by restricting the predictor ranges available to training.

What this establishes. Not a tuning failure — a sample-size floor. Four training years give the model no basis for a fifth. The 2024–2028 trajectories are therefore projections under stated assumptions, with no validated predictive skill, and are reported as illustrations rather than forecasts.

Changes from v3.0 and corrections log

v3.1 restructured the GPR so management variables can carry effects, and a verification pass against the executed notebook corrected several reported values.

Model changes

Change	v3.0	v3.1	Why
GPR structure	18 fits, one per treatment × grazing × depth	One global fit, n = 264	Treatment, grazing, and depth were constants inside every v3.0 fit
Kernel	Isotropic Matérn	ARD Matérn (one length-scale per input)	Gives a second, independent relevance ranking
Validation	Leave-one-year-out RF only	Adds 2023 temporal holdout with coverage and width	Tests whether uncertainty bounds are calibrated
SPEI source	Hand-entered annual dictionary	GRIDMET/DROUGHT spei90d per sampling window	Annual values were collinear with year by construction
Forecast SPEI	0.0	Observed mean (≈ -0.6)	0.0 implied wetter than 7 of 9 observed windows
Horizon	2030	2028	Keeps projection length equal to the observed record
Wording	"Forecast"	"Projection under stated assumptions"	No validated predictive skill

Corrections

Item	Error	Correction
SPEI placeholder	The dictionary that ran in v2.1–v3.0 held invented values (2021 = -2.10 , 2022 = $+0.88$) that contradict both the manual SPEI pull and GEE	Replaced by GEE extraction. The manually pulled values agree with GEE in direction; the error was in the placeholder, not the source data
GPR feature selector	year listed twice, giving 11 columns for 10 feature names	Deduplicated with <code>dict.fromkeys</code>
Tableau export	spei_df overwritten by the placeholder; RF importance values misaligned to feature names	Separate variable; importances reindexed to RF_FEATURES

Item	Error	Correction
Poster, panel 3	Legume NO_3^- values shifted one column (+1.79 shown under 2020)	Moved to 2021; 2020 left blank (not sampled)
Poster, panel 2	ARD bars on a separate scale from MDI bars	Both on one share scale, max = NDVI ARD 0.320
Poster, panel 2	Grazing described as "last under permutation"	Ranks 9th of 11, in the negative-importance group
T1 / T2 labels	Read as Treatment 1 / 2 in the study table	Mean Time 1 / 2. Carbon fractions, FAME, and pH/EC cover all treatments at Time 2 in 2022–2023
Abstract	Claimed carbon fractions and FAME were restricted to Fertilized Triticale	Clause removed

Model outputs were checked against the executed v3.1 notebook and `grazing_stats_v3.csv`; sample counts were computed from `NH4_NO3_master`.

Limitations

The binding limit is time: five annual campaigns give every method fewer independent observations than its rankings imply.

- **Effective sample size.** 264 modelling rows are 9 plots $\times$ 5 years $\times$ 2 depths $\times$ 2 grazing conditions. Temporally there are five independent points, and year-level covariates like SPEI have nine values in total.
- **Feature selection is not nested.** RF ranked covariates on all data before the GPR was fit and validated, so the GPR's uncertainty is optimistic (Ambroise & McLachlan, 2002).
- **Plot is not a grouping unit in cross-validation.** Repeated measures on fixed plots mean rows are not independent. Folds were grouped by year, not by plot.
- **SPEI is site-level.** One 4.6 km GRIDMET pixel covers the field, so water balance cannot vary between plots and remains partly confounded with year ($r = -0.24$).
- **GHG coverage is asymmetric.** Flux sampling ends in August, so no flux value aligns with Time 2 (September) nitrogen. GHG enters as a seasonal mean, not a matched measurement.
- **Unbalanced grazing samples.** Grazed outnumber ungrazed 2:1 to 3.5:1. Mann–Whitney tolerates unequal groups, but power is set by the smaller group.
- **Multiple comparisons — resolved.** 2023 is the primary contrast; results are Holm-corrected within six tests and reported at $p < 0.05$ rather than $p < 0.01$.
- **Outlier handling.** NH_4^+ and NO_3^- were capped at the 90th percentile for modelling, and seven CH_4 values were blanked by judgement. Both choices shape the fitted surface and should be reported.

Next steps

Each step targets one limit above; the first two are analysis changes that need no new data.

- ☐ **Nest the feature selection.** Rank covariates inside each training fold, never on the full dataset, so the GPR's uncertainty reflects selection error.
- ☐ **Group folds by plot as well as year.** Repeated measures on fixed plots should never straddle train and test.
- ☐ **State the correction family for the grazing test.** Declare 2023 as the primary contrast (Holm, 6 tests) or report that the result does not survive correction across all 22 cells.
- ☐ **Extend the record and rerun the holdout each season.** The holdout, not a fixed threshold, will show when the record can carry a forecast.
- ☐ **Measure moisture at plot resolution.** In-situ probes or plot-scale SAR would let water balance vary between plots and separate it from elapsed time.
- ☐ **Add Time 1 sampling for carbon fractions and FAME.** Both cover all treatments in 2022–2023 at Time 2 only; a second time point and further seasons would turn a snapshot into a trajectory.

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Methods

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Domain

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